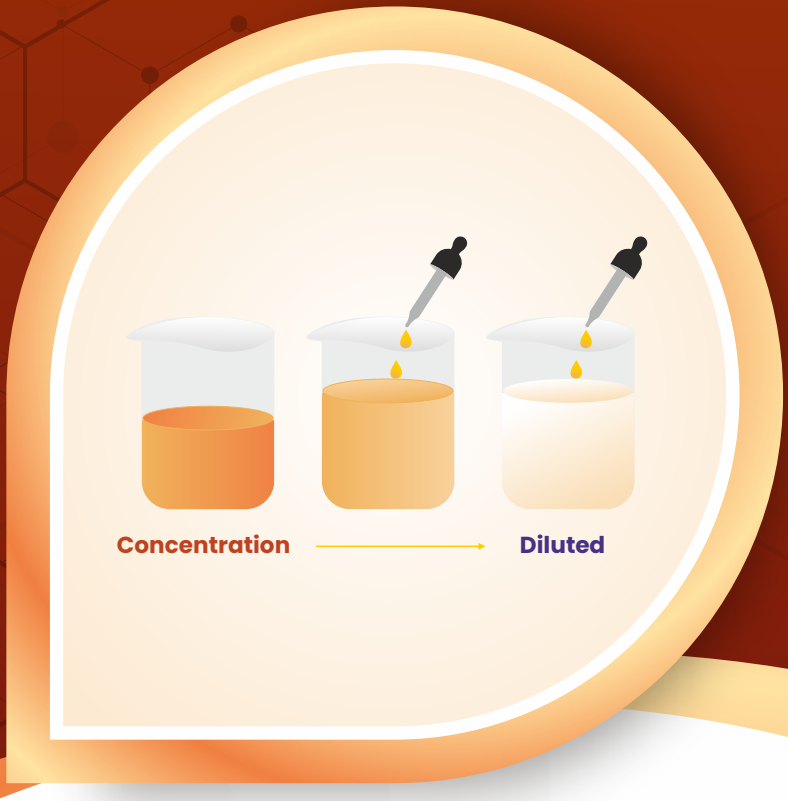
ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

path to success

PHYSICAL CHEMISTRY

JEE (MAIN+ADVANCED)

NURTURE COURSE



Exercise

Concentration Terms

(English Medium)

EXERCISE # S-I

CONCENTRATION TERMS

1. (a) Calculate number of milimoles of 4.9 g H_2SO_4 **CT0115**

(b) 50 mL, 0.2 M NaOH is mixed with 20 mL, 0.1 M NaOH. Calculate number of milimoles of Na^+ in final solution. **CT0115**

(c) 3.65 g HCl, 9.80 g H_2SO_4 and 4.9 g H_3PO_4 are mixed together. Calculate the total milimoles of H^+ in the final solution. **CT0115**

(d) 10 mL, 0.1 M NaOH and 10 mL, 0.2 M $Ca(OH)_2$ are added to 80 mL water. Calculate final milimoles & molarity of $[OH^-]$ in final solution. **CT0115**

2. Calculate the molarity of the following solutions :

(a) 4g of caustic (NaOH) soda is dissolved in 200 mL of the solution. **CT0001**

(b) 5.3 g of anhydrous sodium (Na_2CO_3) carbonate is dissolved in 100 mL of solution. **CT0001**

(c) 0.365 g of pure HCl gas is dissolved in 50 mL of solution. **CT0001**

3. Density of a solution containing 13% by mass of sulphuric acid is 0.98 g/mL. Then molarity of solution will be **CT0002**

4. 15 g of methyl alcohol is present in 100 mL of solution. If density of solution is 0.90 g mL^{-1}, calculate the mass percentage of methyl alcohol in solution **CT0003**

5. Units of parts per million (ppm) or per billion (ppb) are often used to describe the concentrations of solutes in very dilute solutions. The units are defined as the number of grams of solute per million or per billion grams of solvent. Bay of Bengal has 2.1 ppm of lithium ions. If the molality of Li^+ is $x \times 10^{-4}$ m, then find x. (Li = 7) **CT0004**

6. A 7.0 M solution of KOH in water contains 28% by mass of KOH. What is density of solution in gm/ml ? (K = 39) **CT0005**

7. The concentration of a solution is 8% (w/w) and 10% (w/v). Calculate density (in gm/mℓ) of solution? **CT0006**

8. The mole fraction of solute in aqueous urea solution is 0.2. Calculate the mass percent of solute ? **CT0007**

9. The concentration of $Ca(HCO_3)_2$ in a sample of hard water is 405 ppm. The density of water sample is 1.0 gm/ml. If the molarity of solution is $x \times 10^{-3}$ M then find x ? **CT0008**

10. How much $BaCl_2$ (in gm) would be needed to make 250 ml of a solution having the same concentration of Cl^- as one containing 1.825 gm HCl per 100 ml ? (Ba = 137) **CT0009**

11. Calculate **molality (m)** of each ion present in the aqueous solution of **2M NH_4Cl** assuming 100% dissociation according to reaction.

$$NH_4Cl\,(aq) \longrightarrow NH_4^+\,(aq) + Cl^-\,(aq)$$

Given : Density of solution = 3.107 gm / ml. **CT0010**

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PROBLEMS RELATED WITH MIXING & DILUTION

12. Find out the volume (L) of 98% w/w H_2SO_4 (density = 1.8 gm/ ml), must be diluted to prepare 12.6 litres of 2.0 M sulphuric acid solution. **CT0011**

13. 500 ml of 2 M NaCl solution was mixed with 200 ml of 1/4 M NaCl solution. Calculate the molarity of NaCl in final solution. **CT0012**

14. A mixture containing equimolar amounts of $Ca(OH)_2$ and $Al(OH)_3$ requires 0.5 L of 4.0 M HCl to react with it completely. Total moles of the mixture is : **CT0013**

15. 500 gm of urea solution of mole fraction 0.2 is diluted to 1500 gm. Calculate the mole fraction of solute in the diluted solution ? **CT0014**

16. When V ml of 2.2 M H_2SO_4 solution is mixed with 10 V ml of water, the volume contraction of 2% take place. Calculate the molarity of diluted solution ? **CT0015**

17. What volume (in mℓ) of 0.8 M $AlCl_3$ solution should be mixed with 50 ml of 0.2M $CaCl_2$ solution to get solution of chloride ion concentration equal to 0.6 M ? **CT0016**

18. A solution containing 200 ml 0.5 M KCl is mixed with 50 ml 19% w/v $MgCl_2$ and resulting solution is diluted 8 times. Molarity of chloride ion is final solution is : **CT0017**

VOLUME STRENGTH

19. 500 ml of a H_2O_2 solution on complete decomposition produces 2 moles of H_2O. Calculate the volume strength of H_2O_2 solution? **CT0020**

20. The volume strength of 100 ml H_2O_2 solution which produce 5.6 litre of oxygen gas at 1 bar & 0ºC. **CT0021**

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EXERCISE # S-II

1. What volume of 0.2 M NaOH (in L) solution should be mixed to 500 ml of 0.5 M NaOH solution so that 300 ml of final solution is completely neutralised by 20 ml of 2 M H_3PO_4 solution.

CT0022

2. How much minimum volume (in L) of $\left(\frac{5}{51}\right)$ M aluminium sulphate solution should be added to excess calcium nitrate to obtain atleast 1 gm of each salt in the reaction.

$Al_2(SO_4)_3 + 3Ca(NO_3)_2 \longrightarrow 2Al(NO_3)_3 + 3CaSO_4$ **CT0023**

3. One litre of milk weighs 1.035 kg. The butter fat is 4% (v/v) of milk and has density of 875 kg/m^3. If the density of fat free skimed milk is 'x' gm/ml, the value of (x) is ? **CT0024**

4. 100 ml of 0.1 M solution of AB (d = 1.5 gm/ml) is mixed with 100 ml of 0.2 M solution of CB_2 (d = 2.5 gm/ml). Calculate the molarity of B^- in final solution if the density of final solution is 4 gm/ml. Assuming AB and CB_2 are non reacting & dissociates completely into A^+, B^-, C^{+2}.

CT0025

5. 60 ml of a **"x"** % w/w alcohol by weight (d = 0.6 g/cm^3) must be used to prepare 200 cm^3 of 12% alcohol by weight (d = 0.90 g/cm^3). Calculate the mass of alcohol (in gm) in original sample.

CT0026

6. If 0.5 M methanol undergo self dissociation like $CH_3OH \rightleftharpoons CH_3O^- + H^+$ & if concentration of H^+ is 2.5×10^{-4} M then calculate % dissociation of methanol. **CT0027**

7. Determine the volume (in L) of diluted nitric acid (d = 1.11 g mL^{-1}, 20% w/v HNO_3) that can be prepared by diluting 50 mL of conc. HNO_3 with water (d =1.42 g mL^{-1}, 70% w /v). **CT0028**

8. 50 ml of '20V' H_2O_2 is mixed with 200 ml, '10V' H_2O_2. The volume strength of resulting solution is **CT0029**

9. 500 ml of 0.90M CH_3COOH solution is mixed with 600 ml 12% w/v CH_3COOH solution then calculate the final molarity of solution. **CT0030**

10. 45.4 V H_2O_2 solution (500 ml) when exposed to atmosphere looses 11.2 litre of O_2 at 1 atm, & 273 K. New molarity of H_2O_2 solution (Assume no change in volume) **CT0031**

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EXERCISE # O-I

CONCENTRATION TERMS

1. 125 ml of 8% w/w NaOH solution (sp. gravity 1) is added to 125 ml of 10% w/v HCl solution. The nature of resultant solution would be _____.

(A) Acidic (B) Basic (C) Neutral (D) Can not be predicted **CT0032**

2. The molarity of pure water is :

(A) 100 M (B) 55.6 M (C) 50 M (D) 18M **CT0033**

3. Mole fraction of $C_3H_5(OH)_3$ (glycerine) in a solution of 36 g of water and 46 g of glycerine is :

(A) 0.46 (B) 0.36 (C) 0.20 (D) 0.40 **CT0034**

4. A molal solution is one that contains one mole of a solute in

(A) 1000 g of the solvent (B) one litre of the solution

(C) one litre of the solvent (D) 22.4 litres of the solution **CT0035**

5. The molarity of a solution of sodium chloride in water containg 5.85 gm of sodium chloride in 500 ml of solution is :-

(A) 0.25 M (B) 2.0 M (C) 1.0 M (D) 0.2 M **CT0036**

6. The molarity of 98% by wt. H_2SO_4 (d = 1.8 g/ml) is

(A) 6 M (B) 18 M (B) 10 M (D) 4 M **CT0037**

7. Which one of the following modes of expressing concentration of solution is independent of temperature -

(A) Molarity (B) Molality (C) % w/v (D) Grams per litre **CT0038**

8. Equal weight of NaCl and KCl are dissolved separately in equal volumes of solutions. Molarity of the solutions will be –

(A) Equal (B) Greater for NaCl

(C) Greater for KCl (D) Uncomparable. **CT0039**

9. Molality of 20% (w/w) aq.glucose solution is :

(A) $\frac{25}{18}$ m (B) $\frac{10}{9}$ m (C) $\frac{25}{9}$ m (D) $\frac{5}{18}$ m **CT0042**

10. Molarity of liquid HCl, if density is 1.17 g/cc. :

(A) 36.5 M (B) 18.25 M (C) 32.05 M (D) 42.10 M **CT0043**

11. Molarity and Molality of a solute (M. wt = 50) in aqueous solution is 9 and 18 respectively. What is the density of solution.

(A) 1 g/cc (B) 0.95 g/cc (C) 1.05 g/cc (D) 2 g/cc **CT0093**

12. The relationship between mole fraction (X_A) of the solute & molality 'm' of its solution in ammonia would be

(A) $\frac{55.56(X_A)}{1-X_A}$ =m (B) $\frac{58.82(X_A)}{1-X_A}$ = m (C) $\frac{58.82(1-X_A)}{X_A}$ =m (D) $\frac{55.56(1-X_A)}{X_A}$ =m

CT0052

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13. 3.0 molal NaOH solution has a density of 1.12 g/mL. The molarity of the solution is-

(A) 2.97 M (B) 3 M (C) 3.05 M (D) 3.5 M **CT0053**

14. The concentrated sulphuric acid that is peddled commercially is 95% H_2SO_4 by weight. If the density of this commerical acid is 1.834 g cm^{-3}, the molarity of this solution is :-

(A) 17.8 M (B) 15.7 M (C) 10.5 M (D) 12.0 M **CT0087**

15. The density of a solution prepared by dissolving 120 g of urea (mol. mass = 60 u) in 1000 g of water is 1.15 g/mL. The molarity of this solution is

(A) 2.05 M (B) 0.50 M (C) 1.78 M (D) 1.02 M **CT0088**

16. A 5.2 molal aqueous solution of methyl alcohol, CH_3OH, is supplied. What is the mole fraction of methyl alcohol in the solution ?

(A) 0.086 (B) 0.050 (C) 0.100 (D) 0.190 **CT0089**

PROBLEMS RELATED WITH MIXING AND DILUTION

17. How much water should be added to 200 cc of semimolar solution of NaOH to make it exactly decimolar?

(A) 1000 cc (B) 400 cc (C) 800 cc (D) 600 cc **CT0040**

18. 100 ml of 0.3 M HCl solution is mixed with 200 ml of 0.3 M H_2SO_4 solution. What is the molarity of H^+ in resultant solution ?

(A) 0.9 M (B) 0.6 M (C) 0.4 M (D) 0.5 M **CT0041**

19. The molarity of a solution made by mixing 50 ml of conc. H_2SO_4 (18 M) with 50 ml of water, is:

(A) 36 M (B) 18 M (C) 9 M (D) 6M **CT0044**

20. Equal volumes of 10% (w/v) of HCl is mixed with 10% (w/v) NaOH solution. The resultant solution be.

(A) basic (B) neutral (C) acidic (D) can't be predicted.

CT0045

21. What volume of 0.2M NaOH solution is needed for complete neutralisation of 0.49 gm orthophosphoric acid -

(A) 75 ml (B) 300 ml (C) 0.075 ml (D) 50 ml **CT0046**

22. Assuming complete precipitation of AgCl, calculate the sum of the molar concentration of all the ions if 2 lit of 2M Ag_2SO_4 is mixed with 4 lit of 1 M NaCl solution is :

(A) 4M (B) 2M (C) 3 M (D) 2.5 M **CT0050**

VOLUME STRENGTH

23. 20 ml of '20 vol' H_2O_2 solution is diluted to 80 ml. The final volume strength of solution is

(A) '80 vol' (B) '25 vol' (C) '5 vol' (D) '8 vol' **CT0096**

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EXERCISE # O-II

ASSERTION/REASON :

1. **Statement -1** : Molality of pure ethanol is lesser than pure water.

Statement -2 : Density of ethanol is lesser than density of water.

[Given : $d_{ethanol}$ = 0.789 gm/ml; d_{water} = 1 gm/ml]

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is false, statement-2 is true.

(D) Statement-1 is true, statement-2 is false. **CT0054**

2. **Statement-1 :** Molarity and molality have almost same value for a very dilute aqueous solution.

Statement-2 : In all very dilute solution, the mass of solvent (in gm_) is equal to the volume of solution (in ml).

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true. **CT0055**

3. **Statement-1 :** The mass fraction of solute in a solution is always greater than its mole fraction.

Statement-2 : Mole fraction of solvent in an aqueous solution of ethanol must be greater than that of solute.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true. **CT0056**

ONE OR MORE THAN ONE CORRECT :

4. The molar concentration of HCl (aq.) is 10^{-5} M. Which of the following statements are correct.

($d_{solution}$ = 1 gm /cc)

(A) The mole fraction of HCl $\cong 1.8 \times 10^{-7}$

(B) The concentration of HCl in ppm is 3.65 ppm

(C) The molality of HCl solution is approximately 10^{-5} m

(D) The (w/v)% of solution is 3.65×10^{-5} % **CT0057**

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5. Solution(s) containing 40 gm NaOH is/are

(A) 50 gm of 80% (w/w) NaOH

(B) 50 gm of 80% (w/v) NaOH [$d_{soln.}$ = 1.2 gm/ml]

(C) 50 gm of 20 M NaOH [$d_{soln.}$ = 1 gm/ml]

(D) 50 gm of 5m NaOH **CT0058**

6. The **incorrect** statement(s) regarding 2M $MgCl_2$ aqueous solution is/are ($d_{solution}$ = 1.09 gm/ml)

(A) Molality of Cl^- is **4.44 m** (B) Mole fraction of $MgCl_2$ is exactly **0.035**

(C) The conc. of $MgCl_2$ is **19% w/v** (D) The conc. of $MgCl_2$ is **19×10^4 ppm**

CT0059

7. A sample of H_2O_2 solution labelled as 56.75 volume has density of 530 gm/L. Mark the correct option(s) representing concentration of same solution in other units. (Solution contains only H_2O and H_2O_2)

(A) $M_{H_2O_2} = 6$ (B) $\% \frac{w}{v} = 17$

(C) Mole fraction of $H_2O_2 = 0.25$ (D) $m_{H_2O_2} = \frac{1000}{72}$ **CT0060**

8. 100 mL of 0.06 M $Ca(NO_3)_2$ is added to 50 mL of 0.06 M $Na_2C_2O_4$. After the reaction is complete (CaC_2O_4 is precipitated)

(A) 0.003 moles of calcium oxalate will get precipitated

(B) 0.003 M Ca^{2+} will remain in excess

(C) $Na_2C_2O_4$ is the limiting reagent

(D) Oxalate ion ($C_2O_4^{2-}$) concentration in final solution is 0.003 M **CT0061**

PARAGRAPH BASED :

Comprehension Q.9 and Q.10 (2 questions)

2 litre of 9.8 % w/w H_2SO_4 (d = 1.5 gm/ml) solution is mixed with 3 litre of 1 M KOH solution.

9. The number of moles H_2SO_4 added are

(A) 1 (B) 2 (C) 3 (D) 0.5 **CT0062**

10. The concentration (in molarity) of H^+ if solution is acidic or concentration of OH^- if solution is basic in the final solution is

(A) 0 (B) $\frac{3}{10}$ (C) $\frac{3}{5}$ (D) $\frac{2}{5}$ **CT0063**

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Comprehension Q.11 and Q.14 (4 questions)

Estimation of nitrogen : There are two methods for the estimation of nitrogen (i) Dumas method and (ii) Kjedahl's method.

i. **Dumas method :** A known mass of compound is heated with copper oxide (CuO) in an atomsphere of CO_2, which gives free nitrogen along with CO_2 and H_2O.

$C_xH_yN_z + (2x + y/2)\ CuO \rightarrow xCO_2 + y/2\ (H_2O) + z/2\ (N_2) + (2x + y/2)\ Cu.$

The gaseous mixture is passed over a heated copper gauze which converts traces of nitrogen oxides formed to N_2. The gaseous mixture is collected over an aqueous solution of KOH which absorbs CO_2, and nitrogen is collected in the upper part of the graduated tube.

ii. **Kjeldahl's method :** A known mass of organic compound (0.5 gm) is mixed with K_2SO_4 (10 gm) and $CuSO_4$. (1.0 gm) or a drop of mercury (Hg) and conc. H_2SO_4 (25 ml) , and heated in Kjeldahl's flask. $CuSO_4$ or Hg acts as a catalyst, while K_2SO_4 raises the boiling point of H_2SO_4. The nitrogen in the organic compound is quantitatively converted to ammonium sulphate. The resulting mixture is then distilled with excess of NaOH solution and the NH_3 evolved is passed into a known but excess volume of standard HCl or H_2SO_4 . The acid left unused is estimated by titration with some standard alkali. The amount of acid used against NH_3 can thus be known and from this the percentage of nitrogen is calculated.

(a). $C + H + S \xrightarrow[H_2SO_4]{\text{conc.}} CO_2 + H_2O + SO_2$

(b) $N \xrightarrow[H_2SO_4]{\text{conc.}} (NH_4)_2SO_4$

(c) $(NH_4)_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + 2NH_3 + 2H_2O$

(d). $2NH_3 + H_2SO_4 \rightarrow (NH_4)_2SO_4$

iii. This method is not applicable to compounds containing N in nitro and azo groups, and N present in the ring (e.g. , pyridine) as N of these compounds does not change to $(NH_4)_2SO_4$ (ammonium sulphate) under these reaction condtions.

11. 0.30 gm of an organic compound gave 82.1 ml of nitrogen collected at 300K and 775 mm pressure in Dumas method. Calculate the percentage of nitrogen in the compound. (Vapour pressure of water or aqueous tension of water at 300K is 15 mm.

(A) 31.11 (B) 15.56 (C) 28.0 (D) 31.72 **CT0064**

12. 0.50 gm of an organic compound was treated according to Kjeldahl's method. The ammonia evolved was absorbed in 50 ml of 0.5M H_2SO_4. The residual acid required 60 ml of M/2 NaOH solution. Find the percentage of nitrogen in the compound.

(A) 50 (B) 56 (C) 66 (D) 40 **CT0065**

13. 0.4 gm of an organic compound was treated according to Kjeldahl's method. The ammonia evolved was absorbed in 50 ml of 0.5M H_3PO_3. The residual acid required 30 ml of 0.5M $Ca(OH)_2$. Find the percentage of N_2 in the compound.

(A) 20 (B) 50 (C) 70 (D) 45 **CT0066**

14. 0.002 gm of an organic compound was treated according to Kjeldahl's method. 0.2×10^{-4} mol of H_2SO_4 was required to neutralise NH_3. Calculate the percentage of N_2.

(A) 50 (B) 28 (C) 70 (D) 18 **CT0067**

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TABLE TYPE QUESTION

	Column-I		Column-II		Column-III
(A)	2 M - aqueous NaOH solution (density = 1.25 gm/ml)	(i)	2 mole solute/litre solution	(I)	6 % (w/v) solution
(B)	1.5 m - aqueous NaOH solution (density = 1.06 gm/ml)	(ii)	1.5 mole solute/litre solution	(II)	8 % (w/v) solution
(C)	0.5 M aqueous Glucose solution (density = 1.09 gm/ml)	(iii)	0.5 mole solute/litre solution	(III)	9 % (w/v) solution
(D)	1.5 M aqueous Urea solution (density = 1.15 gm/ml)	(iv)	1.5 mole solute/kg solvent	(IV)	9 gm solute per 100 gm solvent

15. Which of the following is incorrect match ?

(A) A – i – II (B) B – ii – I (C) C – iii – IV (D) D – iv – III **CT0068**

16. Which of the following is correct match ?

(A) A – i – IV (B) B – iv – I (C) C – iii – I (D) D – ii – I **CT0068**

17. Which of the following is correct match ?

(A) A – ii – III (B) B – ii – III (C) C – ii – III (D) D – ii – III **CT0068**

MATCH THE COLUMN :

18. Match the column-

Column-I (Concentration of aqueous solution)	**Column-II (Density of given solutions is 1.2 g/ml)**
(A) 2M NaOH solution	(P) 16gm solute in 240gm solution
(B) $8\%\left(\frac{w}{V}\right)$ KOH solution	(Q) 60gm solute in 240 gm solution
(C) $25\%\left(\frac{w}{W}\right)$ $CaCO_3$ solution	(R) 8gm solute in 100 ml solution
(D) $X_{C_3H_7OH} = \frac{1}{11}$	(S) 30 gm solute in 100 ml solution
	(T) 1 mole solute in 400 gm solution **CT0069**

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19. Match the column:

Column I	**Column II**
(A) 20 V H_2O_2	(P) 2.5 M
(B) 24.5 % w/v H_2SO_4	(Q) 1.76 M
(C) Pure water	(R) 1.5 M
(D) 5% w/w NaOH ($d_{solution}$= 1.2 gm/ml)	(S) 55.5 M

CT0070

20.

Column-I	**Column-II**
(A) 120 g CH_3COOH in 1 L solution (d_{sol} = 1.2 g/mL)	(P) M = 2
(B) 120 g glucose dissolved in 1 L solution (d_{sol} = 1.2 g/mL)	(Q) 10% w/w solution
(C) $X_{NH_2CONH_2}$ = 1/31 (aqueous solution)	(R) 12% w/v solution
(D) 19.6% (w/v) H_2SO_4 solution → ($d_{solution}$ = 1.2 g/mL)	(S) m = 1.85
	(T) m = 0.617

CT0071

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EXERCISE # J-MAINS

1. 10 mL of 2(M) NaOH solution is added to 200 mL of 0.5 (M) of NaOH solution. What is the final concentration ? **[JEE(Main-online)-2013]**

(A) 0.57 M (B) 5.7 M (C) 11.4 M (D) 1.14 M

CT0085

2. The density of 3M solution of sodium chloride is 1.252 g mL^{-1}. The molality of the solution will be (molar mass, NaCl = 58.5 g mol^{-1}) **[JEE(Main-online)-2013]**

(A) 2.18 m (B) 3.00 m (C) 2.60 m (D) 2.79 m **CT0086**

3. The amount of $BaSO_4$ formed upon mixing 100 mL of 20.8% $BaCl_2$ solution with 50 mL of 9.8% H_2SO_4 solution will be : **[JEE(Main-online)-2014]**

(Ba = l37, Cl = 35.5, S=32, H = 1 and O = 16)

(A) 33.2 g (B) 11.65 g (C) 23.3 g (D) 30.6 g **CT0083**

4. For the estimation of nitrogen, 1.4 g of an organic compound was digested by Kjeldahl method and the evolved ammonia was absorbed in 60 mL of $\frac{M}{10}$ sulphuric acid. The unreacted acid required 20 mL of $\frac{M}{10}$ sodium hydroxide for complete neutralizaton. The percentage of nitrogen in the compound is : **[JEE(Main-online)-2014]**

(A) 3% (B) 5% (C) 6% (D) 10% **CT0084**

5. The amount of sugar ($C_{12}H_{22}O_{11}$) required to prepare 2 L of its 0.1 M aqueous solution is : **[JEE-Main(Jan.)-2019]**

(A) 68.4 g (B) 17.1 g (C) 34.2 g (D)136.8 g **CT0076**

6. A solution of sodium sulfate contains 92 g of Na^+ ions per kilogram of water. The molality of Na^+ ions in that solution in mol kg^{-1} is: **[JEE-Main(Jan.)-2019]**

(A) 16 (B) 8 (C) 4 (D) 12 **CT0077**

7. 8g of NaOH is dissolved in 18g of H_2O. Mole fraction of NaOH in solution and molality (in mol kg^{-1}) of the solutions respectively are : **[JEE-Main(Jan.)-2019]**

(A) 0.167, 11.11 (B) 0.2, 22.20 (C) 0.2, 11.11 (D) 0.167,22.20 **CT0078**

8. The volume strength of 1M H_2O_2 is: (Molar mass of H_2O_2 = 34 g mol^{-1}) **[JEE-Main(Jan.)-2019]**

(A) 16.8 (B) 11.35 (C) 22.4 (D) 5.6 **CT0079**

9. The strength of 11.2 volume solution of H_2O_2 is :

[Given that molar mass of H = 1 g mol^{-1} and O = 16 g mol^{-1}] **[JEE-Main(april)-2019]**

(A) 13.6% (B) 3.4% (C) 34% (D) 1.7% **CT0080**

10. What would be the molality of 20% (mass/mass) aqueous solution of KI?

(molar mass of KI = 166 g mol^{-1}) **[JEE-Main(april)-2019]**

(A) 1.08 (B) 1.48 (C) 1.51 (D) 1.35 **CT0081**

11. The mole fraction of a solvent in aqueous solution of a solute is 0.8. The molality (in mol kg^{-1}) of the aqueous solution is **[JEE-Main(april)-2019]**

(A) 13.88×10^{-1} (B) 13.88×10^{-2} (C) 13.88 (D) 13.88×10^{-3} **CT0082**

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12. The molarity of HNO_3 in a sample which has density 1.4 g/mL and mass percentage of 63% is (Molecular Weight of HNO_3 = 63) **[JEE-Main(Jan)-2020]**

CT0072

13. 10.30 mg of O_2 is dissolved into a liter of sea water of density 1.03 g/mL. The concentration of O_2 in ppm is **[JEE-Main(Jan)-2020]**

CT0073

14. The ammonia (NH_3) released on quantitative reaction of 0.6 g urea (NH_2CONH_2) with sodium hydroxide (NaOH) can be neutralized by : **[JEE-Main(Jan)-2020]**

(A) 100 ml of 0.1 M HCl (B) 200 ml of 0.4 M HCl

(C) 100 ml of 0.2 M HCl (D) 200 ml of 0.2 M HCl **CT0074**

15. Ferrous sulphate heptahydrate is used to fortify foods with iron. The amount (in grams) of the salt required to achieve 10 ppm of iron in 100 kg of wheat is _____. **[JEE-Main(Jan)-2020]**

Atomic weight : Fe = 55.85 ; S = 32.0 ; O = 16.00 **CT0075**

16. The volume strength of 8.9 M H_2O_2 solution calculated at 273 K and 1 atm is _____. (R=0.0821 L atm K^{-1} mol^{-1}) (rounded off to the nearest integer) **[JEE(Main-online)-2020]**

CT0097

17. 6.023×10^{22} molecules are present in 10 g of a substance 'x'. The molarity of a solution containing 5 g of substance 'x' in 2 L solution is _____ $\times 10^{-3}$. **[JEE(Main-online)-2020]**

CT0098

18. The strengths of 5.6 volume hydrogen peroxide (of density 1 g/mL) in terms of mass percentage and molarity (M), respectively, are: **[JEE(Main-online)-2020]**

(Take molar mass of hydrogen peroxide as 34 g/mol)

(1) 1.7 and 0.25 (2) 1.7 and 0.5 (3) 0.85 and 0.5 (4) 0.85 and 0.25 **CT0099**

19. 4.5 g of compound A (MW = 90) was used to make 250 mL of its aqueous solution. The molarity of the solution in M is $x \times 10^{-1}$. The value of x is _____. (Rounded off to the nearest integer) **[JEE(Main-online)-2021]**

CT0100

20. The $NaNO_3$ weighed out to make 50 mL of an aqueous solution containing 70.0 mg Na^+ per mL is _____g. (Rounded off to the nearest integer) **[JEE(Main-online)-2021]**

[Given : Atomic weight in g mol^{-1} – Na : 23 ; N : 14 ; O : 16] **CT0101**

21. A 6.50 molal solution of KOH (aq.) has a density of 1.89 g cm^{-3}. The molarity of the solution is _____ mol dm^{-3}. (Round off to the Nearest Integer). **[JEE(Main-online)-2021]**

[Atomic masses: K :39.0 u; O :16.0 u; H :1.0 u] **CT0102**

22. The mole fraction of a solute in a 100 molal aqueous solution _____ $\times 10^{-2}$.

(Round off to the Nearest Integer). **[JEE(Main-online)-2021]**

[Given : Atomic masses : H : 1.0 u, O : 16.0 u] **CT0103**

23. 0.8 g of an organic compound was analysed by Kjeldahl's method for the estimation of nitrogen. If the percentage of nitrogen in the compound was found to be 42%, then _____ mL of 1 M H_2SO_4 would have been neutralized by the ammonia evolved during the analysis. **CT0105**

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24. The density of NaOH solution is 1.2 g cm^{-3}. The molality of this solution is _____ m.
(Round off to the Nearest Integer) **[JEE(Main-online)-2021]**
[Use : Atomic masses : Na : 23.0 u O : 16.0 u, H : 1.0 u
Density of H_2O : 1.0 g cm^{-3}] **CT0106**

25. An aqueous KCl solution of density 1.20 g mL^{-1} has a molality of 3.30 mol kg^{-1}. The molarity of the solution in mol L^{-1} is _____ (Nearest integer) **[JEE(Main-online)-2021]**
[Molar mass of KCl = 74.5] **CT0107**

26. 100 mL of Na_3PO_4 solution contains 3.45 g of sodium. The molarity of the solution is ___ $\times 10^{-2}$ mol L^{-1} . (Nearest integer) **[JEE(Main-online)-2021]**
[Atomic Masses - Na : 23.0 u, O : 16.0 u, P : 31.0 u] **CT0108**

27. The molarity of the solution prepared by dissolving 6.3 g of oxalic acid ($H_2C_2O_4.2H_2O$) in 250 mL of water in mol L^{-1} is x $\times 10^{-2}$. The value of x is _____. (Nearest integer)
[Atomic mass : H : 1.0, C : 12.0, O : 16.0] **[JEE(Main-online)-2021]**
CT0109

28. Sodium oxide reacts with water to produce sodium hydroxide. 20.0 g of sodium oxide is dissolved in 500 mL of water. Neglecting the change in volume, the concentration of the resulting NaOH solution is _____ $\times 10^{-1}$ M. (Nearest integer)
[Atomic mass : Na = 23.0, O = 16.0, H = 1.0] **[JEE(Main-online)-2021]**
CT0110

29. If 80 g of copper sulphate $CuSO_4 \cdot 5H_2O$ is dissolved in deionised water to make 5 L of solution. The concentration of the copper sulphate solution is x $\times 10^{-3}$ mol L^{-1}. The value of x is _____.
[Atomic masses Cu : 63.54 u, S : 32 u, O : 16 u, H : 1 u] **[JEE(Main-online)-2021]**
CT0111

30. Chlorophyll extracted from the crushed green leaves was dissolved in water to make 2 L solution of Mg of concentration 48 ppm. The number of atoms of Mg in this solution is x $\times 10^{20}$ atoms. The value of x is_____. (Nearest Integer)
(Given : Atomic mass of Mg is 24 g mol^{-1}, N_A = 6.02 $\times 10^{23}$ mol–1)
[JEE(Main-online) July-2022]
CT0116

31. When 800 mL of 0.5 M nitric acid is heated in a beaker, its volume is reduced to half and 11.5 g of nitric acid is evaporated. The molarity of the remaining nitric acid solution is x $\times 10^{-2}$ M. (Nearest Integer)
(Molar mass of nitric acid is 63 g mol^{-1}) **[JEE(Main-online) June-2022]**
CT0117

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EXERCISE # JEE-ADVANCED

1. Dissolving 120 g of urea (mol. wt. 60) in 1000 g of water gave a solution of density 1.15 g/mL. The molarity of the solution is **[JEE-2011]**

(A) 1.78 M (B) 2.00 M (C) 2.05 M (D) 2.22 M **CT0092**

2. 29.2 % (w/w) HCl stock solution has a density of 1.25 g mL^{-1}. The molecular weight of HCl is 36.5g mol^{-1}. The volume (mL) of stock solution required to prepare a 200 mL solution of 0.4 M HCl is. **[JEE-2012]**

CT0112

3. A compound $\mathbf{H_2X}$ with molar weight of 80 g is dissolved in a solvent having density of 0.4 g /ml, Assuming no change in volume upon dissolution, the **molality** of a 3.2 molar solution is. **[JEE 2014]**

CT0091

4. The mole fraction of a solute in a solution is 0.1. At 298 K, molarity of this solution is the same as its molality. Density of this solution at 298 K is 2.0 g cm^{-3}. The ratio of the molecular weights of the solute and solvent, $\left(\frac{MW_{solute}}{MW_{solvent}}\right)$, is **[JEE-2016]**

CT0113

5. The mole fraction of urea in an aqueous urea solution containing 900 g of water is 0.05. If the density of the solution is 1.2 g cm^{-3}, the molarity of urea solution is **[JEE 2019]**

(Given data : Molar masses of urea and water are 60 g mol^{-1} and 18 g mol^{-1}, respectively)

CT0090

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ANSWER KEY

EXERCISE # S-I

1. **(a) 50 milimoles, (b) 12 milimoles, (c) 450 milimoles, (d) 5 milimoles, 0.05 M**

2. **(a) 0.50 M, (b) 0.50 M, (c) 0.20 M**

3.	**1.30 M**	**4. 16.66 or 16.67**	**5. 3.00**	**6. 1.40**
7.	**1.25**	**8. 45.45**	**9. 2.50**	**10. 13.00 gm**
11.	**0.66 or 0.67**	**12. 1.40**	**13. 1.50 M**	**14. (0.80)**
15.	**0.05**	**16. 0.20 M**	**17. 5.55 or 5.56**	**18. (0.15 M)**
19.	**45.40 V**	**20. (56.00 V)**		

EXERCISE # S-II

1.	**0.25**	**2. 0.02**	**3. 1.04**	**4. 0.50 M**
5.	**21.60**	**6. 0.05**	**7. 0.17 or 0.18**	**8. (12.00 V)**
9.	**(1.50 M)**	**10. (2.00 M)**		

EXERCISE # O-I

Que.	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
Ans.	A	B	C	A	D	B	B	B	A	C
Que.	**11**	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
Ans.	B	B	B	A	A	A	C	D	C	C
Que.	**21**	**22**	**23**							
Ans.	A	B	C							

EXERCISE # O-II

Que.	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
Ans.	B	C	D	A,C,D	A,C	B,D	B,D	A,C	C	C
Que.	**11**	**12**	**13**	**14**	**15**	**16**	**17**			
Ans.	A	B	C	B	D	B	D			
Que.	**18**				**19**					
Ans.	A-P,R	B-P,R	C-Q,S,T	D-S,Q	A-Q	B-P	C-S	D-R		
Que.	**20**									
Ans.	A-P,Q,R,S	B-Q,R,T	C-Q,S	D-P						

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EXERCISE # J-MAINS

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	D	B	D	A	C	A	B	B	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	14.00	10	C	4.95 to 4.97	100	25	B	2	13
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	9	64	12	5	3	50	20	13	64	24
Que.	31									
Ans.	54									

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	
Ans.	C	8	8.00	9	2.98 or 2.99	

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